

OBJECTIVE 3 COMPUTATIONS

For C2C2 Configuration:

1. Calculate the UAV's mass on Mars

- Given:
 - $Weight = 6.71\text{ N}$
 - $Gravity = 3.73\text{ m/s}^2$
- Formula:
 - $W = mg$
 - $m = \frac{W}{g}$
- Solution:
 - $m = \frac{6.71\text{ N}}{3.73\text{ m/s}^2}$
 - $m = 1.8\text{ kg}$

2. Determine the Net Acceleration

- Given:
 - $Thrust = 7.5346\text{ N}$
 - $Weight = 6.71\text{ N}$
- Formula:
 - $F_{net} = Thrust - Weight$
 - $F_{net} = ma_{net}$
 - $a_{net} = \frac{F_{net}}{m}$
- Solution:
 - $F_{net} = 7.5346\text{ N} - 6.71\text{ N}$
 - $F_{net} = 0.8246\text{ N}$
 - $a_{net} = \frac{0.8246\text{ N}}{1.80\text{ kg}}$
 - $a_{net} = 0.458\text{ m/s}^2$

3. Time-to-Climb 24m

- Given:

- $a_{net} = 0.458 \text{ m/s}^2$
- $d = 24\text{m}$

- Formula:

- $d = \frac{1}{2}at^2$

- $t = \sqrt{\frac{2d}{a_{net}}}$

- Solution:

- $t = \sqrt{\frac{2(24\text{m})}{(0.458 \text{ m/s}^2)}}$

- $t_{24} = 10.24\text{s}$ - **time-to-climb 24m**

4. Vertical Velocity to 24m

- Given:

- $u = 0 \text{ m/s}$ - *assume from rest*
- $d = 24 \text{ m}$
- $a_{net} = 0.458 \text{ m/s}^2$

- Formula:

- $v = \sqrt{u^2 + 2ad}$

- Solution:

- $v = \sqrt{0^2 + 2(0.458 \text{ m/s}^2)(24 \text{ m})}$

- $v = 4.69 \text{ m/s}$ - **vertical climb speed to 24m**

For T3T2 Configuration:

1. Calculate the UAV's mass on Mars

- Given:

- $Weight = 6.71\text{ N}$
- $Gravity = 3.73\text{ m/s}^2$

- Formula:

- $W = mg$
- $m = \frac{W}{g}$

- Solution:

- $m = \frac{6.71\text{ N}}{3.73\text{ m/s}^2}$
- $m = 1.8\text{ kg}$

2. Determine the Net Acceleration

- Given:

- $Thrust = 7.5346\text{ N}$
- $Weight = 6.71\text{ N}$

- Formula:

- $F_{net} = Thrust - Weight$
- $F_{net} = ma_{net}$

$$\blacksquare a_{net} = \frac{F_{net}}{m}$$

- Solution:

- $F_{net} = 12.5944 - 6.71\text{ N}$
 - $F_{net} = 5.8844\text{ N}$
- $a_{net} = \frac{5.8844\text{ N}}{1.80\text{ kg}}$
 - $a_{net} = 3.27\text{ m/s}^2$

3. Time to Climb 24 m

- Given:

- $a_{net} = 3.27\text{ m/s}^2$
- $d = 24\text{ m}$

- Formula:

- $d = \frac{1}{2}at^2$

- $t = \sqrt{\frac{2d}{a_{net}}}$

- Solution:

- $t = \sqrt{\frac{2(24\text{ m})}{(3.27\text{ m/s}^2)}}$

- $t_{24} = 3.83\text{ s}$ **time-to-climb 24m**

5. Vertical Velocity to 24m

- Given:

- $u = 0\text{ m}$ – *assume from ground*

- $d = 24\text{ m}$

- $a_{net} = 3.27\text{ m/s}^2$

- Formula:

- $v = \sqrt{u^2 + 2ad}$

- Solution:

- $v = \sqrt{(0)^2 + 2(3.27\text{ m/s}^2)(24\text{ m})}$

- $v = 12.53\text{ m/s}$ - **vertical climb speed to 24m**